MIL-R-22678 (AS)
Amendment 4
12 Aug 1968
Superseding
Amendment 3
28 September 1967

MILITARY SPECIFICATION

RECEIVER-TRANSMITTER RADIO
RT-542/ASQ and RT-542A/ASQ

This amendment forms a part of Military Specification MIL-R-22678 (WEP) dated 1 November 1960 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-R-22678 (WEP) is changed to MIL-R-22678 (AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 1, Title: Add

"and RT-542A/ASQ."

Page 1, paragraph 1.1: Add

"(5) Communication by means of AM Radio Telephone (A-3) when used in conjunction with Juliet-28 equipment.

Page 1, paragraph 1.2: Add

"(12) Juliet-28 (RT-542A/ASQ only) -
(13) Control, Radio Set C-8057/ARC (RT-542A/ASQ only) -

Page 2, paragraph 2.1, after "Invitation for bids": add

"except the dated document indicated below,"

Page 2, paragraph 2.1, under SPECIFICATIONS Military: Add

"MIL-I-6181B (29 May 1953) Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment (See 2.2(2))"

Page 2, paragraph 2.2: Delete and substitute

"2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

FSC-5821

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(2) Specification MIL-I-6181B is available only from Naval Air Systems Command, Washington 25, D. C. Attention: **Avionics Division.**"

Page 2: Add

"3.2.8 Antenna Conducted Interference - Antenna conducted interference shall meet the requirements of Specification MIL-I-6181B except that at not more than 9 of the 18 injection frequencies the antenna conducted interference due to the main receiver injection frequency shall not exceed 300 microvolts."

*Page 5, paragraph 3.3.2: Add

"the sensitivity of the auxiliary audio and the auxiliary guard audio shall be such that when an R-F input of 5 microvolts (open circuit) modulated 30% at 1000 cycles is applied, the signal plus noise-to-noise ratio shall be not less than 3.0 db for the auxiliary audio and not less than 6 db for the auxiliary guard audio."

Page 5: paragraph 3.3.2.1: Delete and substitute

"3.3.2.1 Degraded or Unusable Channels Due to Internal Interference - Certain main receiver channel frequencies may be degraded or unusable due to internally generated interference. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies of 241.1, 241.2, and 275.6 mc. Six unusable channels 242.1, 246.6, 253.3, 287.1, 322.8 and 339.0 mc are permitted which are independent of the guard receiver frequency."

Page 5, paragraph 3.3.3, Line 2: Delete "25" and substitute "250".

Page 6, paragraph 3.3.4: Delete table and substitute

	TOTAL BANDWIDTH		
		RT-542/ASQ	RT-542A/ASQ
Attenuation Below Resonant Response	Guard Receiver	Main Receiver	Main Receiver
6 db	60 Kc min	60 Kc min	80 Kc min
60 db	400 Kc max	200 Kc max	200 Kc max

Page 6, paragraph 3.3.9.1: Add

"the auxiliary guard output shall be 0.15 - 0.45 volt rms when loaded with 600 ohms.

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Page 7, paragraph 3.3.9.3: Delete and substitute

"3.3.9.3 Auxiliary Circuit - The auxiliary circuits shall deliver only signals detected by the variable frequency receiver. When a 1000 - microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range of 70 to 200 cps shall be within +4.5 and -3.0 db and in the range 200 to 7000 cps shall be within +2 and -3 db of the output obtained at 1000 cps. The RT-542A/ASQ shall not vary more than +4 and -4 db from the output obtained at 1000 cps when the modulation frequency is varied from 70 to 20,000 cps."

Page 8, paragraph 3.3.12.1.1: Delete table and substitute

"Frequency Cps	Relative Output in db RT-542/ASQ	RT-542A/ASQ
150	-8 or lower	-5 or lower
300	-3 to -7	+4 to -4
600	+1 to -2	+3 to -3
1,000	0 (ref. level)	0 (ref. level)
3,000	+2 to -2	+2 to -2
6,000	+2 to -2	+3 to -3
10,000	+1 to -5	+4 to -4
15,000	--	+4 to -4
20,000	--	+4 to -4 "

Page 8, paragraph 3.3.12.2: Delete and substitute

"3.3.12.2 Tone Modulation - Not less than 70 percent modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is keyed. The frequency of modulation shall be 1000 cps ± 20%."

*Page 9, paragraph 3.3.16: Add

"Under the above condition the auxiliary audio and the auxiliary guard shall have a signal plus noise-to-noise ratio not less than 20 db."

Page 11, paragraph 3.4.1.2: Delete "8-57/64" and "20-5/22" and substitute respectively "9" and "20-5/32".

Page 11, paragraph 3.4.1.2: Add

"The front panel shall extend an additional 1-1/16 inches below the base plate and the rear guide pins shall extend an addition 11/32 inches beyond the rear of the case."

Page 11, paragraph 3.4.1.3: Delete "44.5" and substitute "45.0".

Page 12, paragraph 3.4.1.5.3: Delete "PT076-10-6S" and substitute "PT07C-10-6S."

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Page 13, paragraph 3.4.1.8.3: Add

"The auxiliary guard output of the RT-542A/ASQ shall be terminated in a 600 ohm load, including means for measuring the audio output voltage."

Page 13, paragraph 3.4.1.8.4, under

"Carbon Microphone Input:" Delete

"Frequency range 150-10,000 cps" and substitute
"Frequency range 150-10,000 cps (RT-542/ASQ)
150-20,000 cps (RT-542A/ASQ)"

"Dynamic Mike Input": Delete

"Frequency range 150-10,000 cps"and substitute
"Frequency range 150-10,000 cps (RT-542/ASQ)
150-20,000 cps (RT-542A/ASQ)

Add Note: 1.0 volt rms open circuit must be supplied when the RT-542A/ASQ is used with Juliet Equipment."

*An asterisk preceding a page number denotes a change to be made to the specification which did not appear in Amendment 3.

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